



The hidden cost of digital aggression: How engaging in cyberbullying facilitates moral disengagement among children through changes in normative beliefs about aggression

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ABSTRACT

Cyberbullying is an increasingly common public health issue worldwide, with serious implications for the development of children and young people. However, most existing studies have treated normative beliefs about aggression (NOBAGs) and moral disengagement (MD) solely as causes of cyberbullying, while few have examined whether the same cyberbullying behaviors may conversely influence the development of NOBAGs and MD. In this study, a cross-lagged panel model (CLPM) and a random-intercept cross-lagged panel model (RI-CLPM) were used to explore the reciprocal, longitudinal relationships between cyberbullying and MD and the mediating effect of NOBAGs. Participants were 1393 children (mean age at enrollment = 9.38, 56.2% boys) from 16 primary schools in Hong Kong, China. The CLPM analyses revealed significant positive relationships between cyberbullying, NOBAGs, and MD, with NOBAGs acting as a mediator. The RI-CLPM analyses separated stable individual differences from changes over time: There were strong positive between-person relationships among cyberbullying, NOBAGs, and MD, but the only within-person effects were unidirectional, with cyberbullying significantly predicting increases in later NOBAGs and in MD but without NOBAGs as a mediator. From a practical perspective, these results suggest that interventions should not only address individual differences in moral-cognitive vulnerabilities but also consider the impact of cyberbullying on the escalation of MD. The findings also demonstrate how RI-CLPM complements conventional CLPM and prevents the overinterpretation of apparent longitudinal mediation effects.

1. Introduction

The ubiquitous, virtual, and often anonymous nature of the internet—combined with sometimes arbitrary moral norms—allows both children and adults to engage in deviant online behavior (Barlett, 2024). Cyberbullying—a version of offline bullying that is conducted in cyberspace—comprises repeated and hostile behaviors that seek to harm vulnerable individuals or groups using information and communications technology (e.g., computers, cell phones, tablets; Huang et al., 2021) and has become a subject of global concern. However, the effect of cyberbullying can go beyond disrupting the online environment to pose significant risks to children's real-world physical, psychological, and social functioning (e.g., emotional disorders, physical bullying) and their moral development (Falla et al., 2023; Khan et al., 2020).

The Health Behavior in School-Aged Children Survey (World Health Organization, 2024) reported that about 12% of adolescents have

engaged in cyberbullying and 15% have been cyberbullied. The problem is even more concerning in China: A systematic review of 63 studies in 11 countries between 2015 and 2019 found that up to 46.3% of children and adolescents in China have engaged in cyberbullying, while 44.5% have been subjected to online harassment (C. Zhu et al., 2021). With widespread mobile connectivity and easy platform access, the barriers to internet access for minors in China are relatively low, even though they have poorer digital literacy skills than both adults and their international peers (Department for the Protection of Youth Rights and Interests & China Internet Network Information Center, 2023). Although cyberbullying is incorporated into the legal framework for safeguarding minors created by the Law of the People's Republic of China on the Protection of Minors, it remains unaddressed in most formal curricula in Chinese schools (Leung et al., 2018). Given the prevalence of cyberbullying and the severity of the associated real-world problems, understanding the psychological mechanisms underlying it is crucial for

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developing effective interventions.

While numerous studies have documented a range of risk factors that are strongly associated with cyberbullying—including sensation seeking, maladaptive behavior, psychological distress, and psychopathy traits (e.g., Kowalski et al., 2014; Xu et al., 2024)—moral disengagement (MD) warrants particular attention as a key mechanism that explains the emergence of all forms of bullying behaviors among young adults (Perren & Gutzwiller-Helfenfinger, 2012). Cyberbullies may employ the cognitive mechanisms underlying MD that are suggested by social cognitive theory (Bandura, 2016)—such as minimizing harm, displacing responsibility, or dehumanizing victims—to manage the cognitive dissonance between their actions and their self-concepts as moral individuals; however, the same theory also suggests that a vicious circle may then develop between MD and cyberbullying (Falla et al., 2023; J. Yang, Li, et al., 2022). The development of moral agency may thus be a dynamic process shaped by individuals' lived experiences and social environments (Bandura, 1986), and while the unidirectional impact of MD on cyberbullying has been extensively researched, the inverse pathway from cyberbullying to MD is less understood. Emerging evidence suggests that engaging in cyberbullying may lead to a progressive desensitization to the moral impact of aggressive behavior as the cyberbully increasingly relies on MD to avoid feelings of guilt or shame, which can in turn reinforce future aggression (Falla et al., 2023; J. Yang, Li, et al., 2022). However, it remains unclear how cyberbullies progressively disengage from moral self-sanctions and persist in their behaviors.

A fuller understanding of the underlying mechanisms of MD from the perspective of social cognitive theory would extend the current theoretical frameworks and help to clarify the negative consequences for the perpetrators' moral development of repeatedly engaging in cyberbullying; it would also, from a more practical viewpoint, inform the development of more targeted interventions aimed at reducing persistent cyberbullying. Using longitudinal data, this study therefore aims to investigate temporal associations between cyberbullying and MD alongside the underlying psychological mechanisms.

1.1. Cyberbullying and moral disengagement

Cyberbullying involves power imbalances, unwarranted aggression, and morally indefensible behavior (Falla et al., 2023) and creates conditions in which harmful behaviors are normalized and often ignored. The perpetrators then selectively use cognitive mechanisms to disengage from moral censure, minimize personal responsibility for the outcomes of their behavior, distance themselves from their own aggression, and ignore their victim's pain through a self-regulatory processes of MD (Bandura et al., 1996). Indeed, the digital world and its features create a context that is particularly conducive to MD (Bandura, 2004, 2016); for instance, the absence of visual and auditory cues means perpetrators are less likely to perceive victims' suffering and experience empathic concern and may even create the illusion that no harm was caused (Perren & Gutzwiller-Helfenfinger, 2012; Wang & Ngai, 2020). Asynchrony can then shield cyberbullies from immediate accountability while persistent online content can result in the victim being repeatedly presented with the original harmful material, intensifying their distress; this may also, in turn, lead to cyberbullying being reframed as a weapon of counterattack, with victims able to morally justify retaliation by positioning themselves as agents of justice who are punishing wrongdoing (W. Li, 2023). Even victims may thus experience a sense of moral detachment in cyberspace that makes it easier to engage in deviant behavior and harder to maintain ethical standards (da Veiga Simão et al., 2018).

Beyond the features of online communication, the sociocultural context may further affect the development and manifestation of cyberbullying and MD. In China, students can experience enormous social and parental pressure for achievement and even perfectionism (Chyu & Chen, 2022), yet overt aggressive behaviors are generally not

permitted by cultural norms that emphasize group harmony and saving face (Y. Li et al., 2012; F. Yang, Jin, et al., 2022). This cultural tension may therefore make the digital environment an appealing alternative outlet.

Ample evidence has already established a positive relationship between MD and cyberbullying (e.g., X. Li, Ang, et al., 2025; X. Li, Shek, et al., 2025; Lo Cricchio et al., 2020; Paciello et al., 2020); MD may therefore function as a risk factor that predisposes individuals toward cyberbullying or—the inverse—the experience of perpetrating cyberbullying may foster the development of MD as a defensive mechanism (Sticca & Perren, 2015). According to Bandura's (1986) social cognitive theory, MD operates under a triadic framework involving mutual, reciprocal relationships between internal personal factors (cognitive, affective, and biological), behavior patterns, and societal factors. Within this framework, MD is a progressive weakening of internal ethical controls under the influence of personal experience and the social environment: People initially maintain restraint against minor ethical violations due to concerns about potential consequences or internal moral conflict, but through external reinforcement or punishment, the acceptability of immoral behavior gradually increases, making it easier to justify and commit more severe misconduct (Bandura, 2016).

Numerous studies have found MD to be a significant and potent antecedent of cyberbullying among various populations (e.g., X. Li, Ang, et al., 2025; X. Li, Shek, et al., 2025; Lo Cricchio et al., 2020; Paciello et al., 2020), but other research has suggested the inverse pattern of cyberbullying leading to greater MD. For instance, Falla et al. (2023) found that engaging in cyberbullying by adolescents predicted their MD strategies a year later, while J. Yang, Li, et al., (2022) observed among a large sample of Chinese adolescents that cyberbullies were more likely to have higher levels of MD; this was further supported by a recent study among European schoolchildren (Dueñas-Casado et al., 2025). Similarly, Jo (2025) found that engaging in more cyberbullying or disclosure leads to greater rationalization of immorality as perpetrators use cognitive justifications to reconcile their harmful behavior with their positive self-images. Sticca and Perren (2015) also suggested that involvement in bullying in early childhood may compromise the development of moral standards, which would then establish a foundation for aggressive behavior throughout adolescence. In short, there appears to be a self-reinforcing cycle with a bidirectional, reciprocal association between MD and antisocial behaviors.

1.2. Normative beliefs about aggression

While a connection between cyberbullying and MD has been theoretically and empirically established (e.g., Kowalski et al., 2014), the underlying cognitive mechanisms remain insufficiently understood. Social information processing theory offers a detailed account of the mechanisms underlying general aggression in young adults (Crick & Dodge, 1994; Dodge & Crick, 1990) and particularly the role of normative beliefs (Hilvert-Bruce & Neill, 2020). The development of such beliefs is a reflection of socialization processes, which include the internalization of social norms, the influences of significant others (e.g., parents, peers), and subjective evaluations (Huesmann & Guerra, 1997). Normative beliefs establish standards for both appropriate and inappropriate actions, which in turn affect social behavior; normative beliefs about aggression (NOBAGs) then represent the extent to which individuals perceive aggression and bullying as appropriate or acceptable (Huesmann & Guerra, 1997). Notably, NOBAGs can impact how individuals interpret social cues, which can lead them to mistakenly perceive neutral interactions as hostile and thus create a greater tendency toward aggressive responses and actions (Smeijers et al., 2020).

The unique characteristics of the internet—including ambiguity, perceived anonymity, and the lack of temporal and spatial restrictions—can weaken one's sense of individuality and personal responsibility, concern for others' views, and behavioral self-control and regulation, making individuals more reliant on their own beliefs to

direct their behavior (Lowry et al., 2016). In this context, NOBAGs play an important role in shaping how individuals interpret and justify aggressive acts; indeed, studies have shown that stronger favorable NOBAGs are associated with a higher likelihood of children and adolescents engaging in cyberbullying (e.g., Ang et al., 2017; X. Li, Shek, et al., 2025; Peled et al., 2019). Notably, however, NOBAGs are not stable but instead generally influenced by contextual factors (e.g., exposure to violent media) and past social experiences (e.g., engaging in aggressive behaviors) that can make it more likely for a person to choose aggressive strategies to solve problems (Ang et al., 2011; H. Zhou, Wei, et al., 2023); cyberbullying behavior may thus actually promote the development of NOBAGs. Although relevant research is limited, Wright and Li (2013) conducted a longitudinal study to test whether cyberbullying facilitates NOBAGs and found a bidirectional relationship between the two constructs. The present study builds upon this to examine the role of cyberbullying in subsequent changes in NOBAGs.

Beyond their direct association with cyberbullying, NOBAGs may also facilitate the development of MD by undermining moral cognition and distorting moral judgment; for example, Jiang et al. (2022) found that NOBAGs led to higher levels of MD, which in turn indirectly increased the risk of school bullying among Chinese youths. The authors suggested that, when a person engages in aggressive behaviors, NOBAGs—which are key individual factors in determining one's information processing mode—can lead to hostile interpretations and can interfere with and create biases in one's moral cognition; they also suggested that the relatively high tolerance for aggression in traditional Chinese education likely fosters the formation of beliefs about the acceptability of aggression, thus further distorting individual moral cognition (Jiang et al., 2022).

Taken together, such findings have established both that cyberbullying strengthens NOBAGs and that NOBAGs influence MD (e.g., Jiang et al., 2022; Wright & Li, 2013). However, there has been little exploration of how cyberbullying behaviors affect moral cognition and how NOBAGs function in the relationship between cyberbullying and its negative moral outcomes. Another goal of this research is to evaluate whether NOBAGs serve as a mechanism connecting cyberbullying and MD.

1.3. Current study

Prior research has identified significant associations between cyberbullying, NOBAGs, and MD, but it is still unclear how cyberbullying and MD mutually affect each other through cognitive factors—including NOBAGs—in the long term. To clarify the dynamic interplay between cyberbullying, NOBAGs, and MD, we applied a random-intercept cross-lagged panel model (RI-CLPM) to three longitudinal waves of data collected from children in Hong Kong. RI-CLPMs examine whether variations in an individual's score in one construct relate to later changes in another and can separate variance into between-person and within-person components for each observed variable across successive waves of data. The between-person variance captures stable differences between individuals, while the within-person variance represents changes within each individual over time. We also used a cross-lagged panel model (CLPM) as a supplementary analysis to verify the temporal dynamics between the variables despite the limitations of such models, which confound within-person and between-person effects. CLPMs are commonly used to analyze dynamic relationships between constructs (Zyphur et al., 2020) and to help detect real relationships between different constructs that vary over time (Rogosa, 1980; Zyphur et al., 2020). For the current study, we proposed four hypotheses:

H1. There are significant, positive, bidirectional cross-lagged effects between cyberbullying and NOBAGs.

H2. There are significant, positive, bidirectional cross-lagged effects between MD and NOBAGs.

H3. Cyberbullying has a reciprocal, positive relationship with MD over time.

H4. NOBAGs mediate the effect of cyberbullying on MD; specifically, cyberbullying increases NOBAGs, which subsequently increase MD.

2. Methods

2.1. Participants and procedure

The Human Subjects Ethics Sub-Committee of the authors' university approved this study. Prior to data collection, authorization was obtained from the participating primary schools alongside parental permissions and student assent. The students were also informed of the voluntary and anonymous nature of their participation. The first data collection (T1) was completed in 2021, the second (T2) in 2022, and the third (T3) in 2023. Data from participants who completed at least one assessment were included in the analysis. A total of 1393 children from 16 primary schools in Hong Kong were enrolled. Sample sizes in the three waves were 1249 at T1 (mean age = 9.38, $SD = 0.57$; 51.0% boys), 1181 at T2 (mean age = 10.54, $SD = 0.60$; 47.5% boys), and 1144 at T3 (mean age = 11.44, $SD = 0.59$; 45.5% boys). The attrition rates from the initial enrollment were 10.34% at T1, 15.22% at T2, and 17.88% at T3. Each assessment lasted around 30 min. The students could discontinue participation at any time without penalty and ask the research assistants questions if anything was unclear. All the information obtained from the participants was treated as strictly confidential.

We first performed Little's Missing Completely at Random test and found that the missingness patterns of the studied variables were not completely random ($\chi^2 = 117.932$, $df = 79$, $p < 0.01$). Chi-squared tests were then conducted to explore the patterns in the missing data (Little & Rubin, 2019); these found a non-significant relationship between missingness and gender ($p > 0.05$) but a significant relationship with school ($p < 0.001$). Independent samples t-tests were then conducted within each school to compare those participants with complete data to those without. The results showed no significant differences in most comparisons ($p > 0.05$), supporting the conclusion that *which* data were missing was random but that *whether* data were missing was dependent on the school. Thus, instead of removing participants with missing data, a full information maximum likelihood approach was taken, which allowed all available information to be included in the modeling process (Graham, 2009).

2.2. Measures

2.2.1. Cyberbullying

The children reported their own cyberbullying behaviors using the Cyberbullying Questionnaire (Ang & Goh, 2010), which comprises nine items (e.g., "I made fun of someone by sending/posting stories, jokes, or pictures about him/her") responded to on a five-point Likert-type scale coded from 1 ("Never") to 5 ("A few times every week"). Higher scores suggest greater prevalence and frequency of cyberbullying. Cronbach's alpha values for the questionnaire were 0.905 at T1, 0.882 at T2, and 0.883 at T3.

2.2.2. Normative beliefs about aggression

The Normative Beliefs About Aggression Scale (Huesmann & Guerra, 1997) was used to measure individual evaluations of whether cyber-relational and cyber-verbal aggressive behaviors are appropriate (Wright & Li, 2013). The scale comprises 12 items (e.g., "It is usually OK to post mean things about someone online"), each assessed using a six-point Likert-type scale coded from 1 ("strongly disagree") to 6 ("strongly agree"). The Cronbach's alpha values were 0.844 at T1, 0.881 at T2, and 0.900 at T3.

2.2.3. Moral disengagement

The Cyberbullying Moral Disengagement Scale captures levels of MD (Bussey et al., 2015). Eight statements regarding online aggression and morality (e.g., “If people give out their passwords to others, they deserve to be cyberbullied”) are responded to using a five-point Likert-type scale from 1 (“Strongly Disagree”) to 5 (“Strongly Agree”). Higher scores demonstrate greater deviation from decent conduct in the digital context. The scale showed excellent reliability, with Cronbach’s alpha values of 0.850 at T1, 0.823 at T2, and 0.861 at T3.

2.3. Data analysis

Means and standard deviations were calculated for all numeric variables using SPSS 29.0, and Pearson’s correlation coefficients were calculated between those variables. We also tested the longitudinal measurement invariance (LMI) of cyberbullying, NOBAGs, and MD using configural invariance, metric invariance, and scalar invariance models with Mplus 8.11 (Muthén & Muthén, 1998–2024). The configural model allowed the indicators to vary without constraints, while the metric model included constraints to ensure that factor loadings were consistent across the constructs over time. In the scalar model, the corresponding indicator intercepts were constrained to be equal over time. LMI would be considered supported if the model fit did not significantly worsen between the models—that is, if the changes in the comparative fit index (ΔCFI) were below 0.010, the changes in the root mean square error of approximation ($\Delta RMSEA$) were below 0.015, and the changes in the standardized root mean square residual ($\Delta SRMR$) were below 0.030, as suggested by Chen (2007).

We then estimated the CLPM and RI-CLPM. The autoregressive paths in the CLPM indicated the stability of the variables over time, while the cross-lagged paths assessed whether the variations in one variable related to later changes in another variable (Kearney, 2018). In the RI-CLPM, between-person variance was modeled from the three waves of data using three random-intercept factors—one each for the cyberbullying, NOBAGs, and MD latent variables. All factor loadings from the random intercepts to their indicators were fixed at 1, and the intercepts were allowed to freely covary in the model. To capture the within-person components, each observed score was regressed on the respective latent factor, with factor loadings fixed at 1 and measurement error variances set to 0. Covariances between the within-person variables were also allowed. In contrast to the classic CLPM, the RI-CLPM’s autoregressive coefficients reflect within-person carry-over effects in the variables over time, while the cross-lagged paths estimate how a deviation from an individual’s score on one variable predicts subsequent deviations in another variable, adjusted for the preceding expected scores. In both models, we used the “indirect” command in Mplus to calculate the effect size for the mediated paths. All models were calculated with maximum likelihood estimation. Gender and school were constrained as time-invariant covariates with the study variables

(Mulder & Hamaker, 2021) to control for the potential and complex influence of gender and socioeconomic conditions.

Four indicators were used to assess the model fit: CFI, Tucker–Lewis index (TLI), RMSEA, and SRMR. A model was considered to have an acceptable fit if the CFI and TLI values were at least 0.90 and the RMSEA and SRMR values did not exceed 0.08 (Hu & Bentler, 1999; Kline, 2010).

3. Results

3.1. Descriptive and correlational analyses

Descriptive statistics and Pearson correlations at and between T1, T2, and T3 are shown in Table 1. The correlations for the same variables between different time points ranged from 0.195 to 0.535 ($p < 0.001$). Correlations were significant and positive at all time points between cyberbullying and NOBAGs ($r = 0.143$ – 0.423 , $p < 0.001$), between cyberbullying and MD ($r = 0.106$ – 0.414 , $p < 0.001$), and between NOBAGs and MD ($r = 0.151$ – 0.494 , $p < 0.001$).

3.2. Longitudinal measurement invariance

We tested the LMI using multiple-group confirmatory factor analysis. We used the item parceling approach to group the NOBAGs items into two parcels to reduce the risk of inflated measurement errors when using multiple items for latent variables. We found small changes in CFI (ΔCFI s from -0.005 to 0.002), TLI (ΔTLI s from -0.003 to 0.004), SRMR ($\Delta SRMR$ s from -0.001 to 0.012), and RMSEA ($\Delta RMSEA$ s from -0.001 to 0.001) between the configural model, the metric model, and the scalar model (see Table 2). LMI was thus supported for each variable in the study.

3.3. Cross-lagged panel model

The CLPM exhibited a good fit ($\chi^2(8) = 31.883$, $p < 0.001$; CFI = 0.988; TLI = 0.917; RMSEA = 0.056, 90% CI = 0.036–0.076; SRMR = 0.024). The autoregressive paths between the T1, T2, and T3 measures of cyberbullying, NOBAGs, and MD were all significant and positive. Significant cross-lagged associations were found between cyberbullying and NOBAGs and between NOBAGs and MD (see Fig. 1). Cyberbullying significantly predicted increases in MD at the next wave of data collection ($\beta_{\text{Cyberbullying}_T1 \rightarrow \text{MD}_T2} = 0.085$, $p < 0.05$; $\beta_{\text{Cyberbullying}_T2 \rightarrow \text{MD}_T3} = 0.112$, $p < 0.01$), but cyberbullying at T1 did not significantly predict MD at T3 ($\beta_{\text{Cyberbullying}_T1 \rightarrow \text{MD}_T3} = -0.020$, $p > 0.05$). Conversely, MD did not significantly predict subsequent changes in cyberbullying ($\beta_{\text{MD}_T1 \rightarrow \text{Cyberbullying}_T2} = -0.021$, $p > 0.05$; $\beta_{\text{MD}_T2 \rightarrow \text{Cyberbullying}_T3} = 0.009$, $p > 0.05$). Further analysis identified a significant mediation effect of NOBAGs at T2 on the relationship between cyberbullying at T1 and MD at T3 (indirect effect: $\beta = 0.025$, $p = 0.001$, 95% CI = 0.010–0.040). All hypotheses except H3 were thus fully supported in the

Table 1
Pearson correlations, means, and standard deviations.

Variables	1	2	3	4	5	6	7	8	9
1.CB_T1	1								
2.CB_T2	0.288***	1							
3.CB_T3	0.256***	0.535***	1						
4.MD_T1	0.401***	0.106***	0.138***	1					
5.MD_T2	0.196***	0.398***	0.268***	0.261***	1				
6.MD_T3	0.151***	0.288***	0.414***	0.244***	0.430***	1			
7.NOAG_T1	0.303***	0.169***	0.143***	0.245***	0.151***	0.166***	1		
8.NOAG_T2	0.268***	0.421***	0.330***	0.188***	0.425***	0.347***	0.264***	1	
9.NOAG_T3	0.220***	0.306***	0.423***	0.159***	0.265***	0.494***	0.195***	0.377***	1
Means	1.19	1.15	1.15	1.82	1.74	1.71	2.17	1.67	1.62
SD	0.49	0.39	0.36	0.82	0.70	0.72	1.00	0.88	0.85

Note. CB = Cyberbullying; MD = Moral disengagement; NOBAG = Normative beliefs about aggression; SD = Standard Deviations; T1 = Time 1; T2 = Time 2; T3 = Time 3.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 2
Longitudinal measurement invariance.

	Models	Overall fit indices						Comparative fit indices				
		χ^2	df	CFI	TLI	SRMR	RMSEA (90% CI)	Comparison	Δ CFI	Δ TLI	Δ SRMR	Δ RMSEA
Cyberbullying	Model 1	523.852***	349	0.948	0.940	0.054	0.023(0.019-0.027)					
	Model 2	535.670***	365	0.950	0.944	0.066	0.022(0.018-0.026)	M2 VS. M1	0.002	0.004	0.012	−0.001
	Model 3	564.935***	381	0.946	0.942	0.065	0.023(0.019-0.027)	M3 VS. M2	−0.004	−0.002	−0.001	0.001
NOBAG	Model 1	976.652***	543	0.968	0.963	0.029	0.030(0.027-0.033)					
	Model 2	989.471***	563	0.969	0.965	0.032	0.030(0.026-0.033)	M2 VS. M1	0.001	0.002	0.003	0.000
	Model 3	1030.262***	583	0.967	0.965	0.032	0.030(0.027-0.033)	M3 VS. M2	−0.002	0.000	0.000	0.000
MD	Model 1	488.925***	216	0.962	0.951	0.032	0.030(0.027-0.034)					
	Model 2	516.080***	230	0.960	0.952	0.035	0.030(0.026-0.033)	M2 VS. M1	−0.002	0.001	0.003	0.000
	Model 3	562.319***	244	0.955	0.949	0.036	0.031(0.027-0.034)	M3 VS. M2	−0.005	−0.003	0.001	0.001

Note. Model 1 = Configural invariance; Model 2 = Metric invariance; Model 3 = Scalar invariance; MD = Moral disengagement; NOBAG = Normative beliefs about aggression; CFI = the comparative fit index; TLI = the Tucker–Lewis index; SRMR = the standardized root mean square residual; RMSEA = the root mean square error of approximation.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

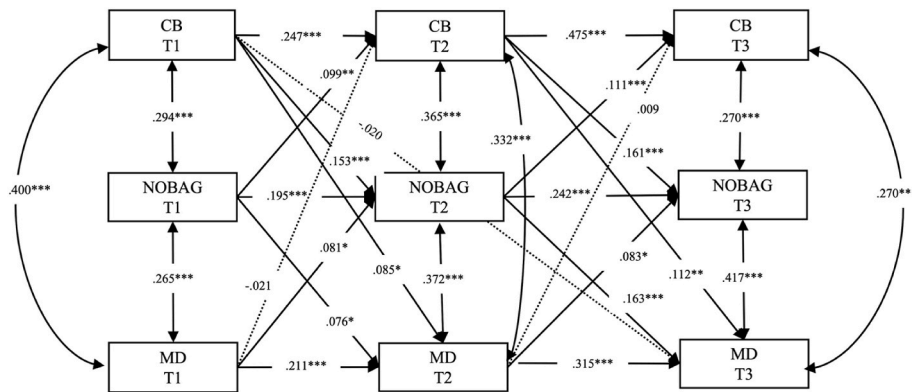


Fig. 1. Cross-lagged panel model of cyberbullying, normative beliefs about aggression, and moral disengagement.

Note. CB = Cyberbullying; MD = Moral disengagement; NOBAG = Normative beliefs about aggression. Rectangles denote observed variables; arrows indicate significant relationships; dashed arrows signify non-significant associations; double-headed arrows illustrate correlations. Horizontal arrows depict autoregressive effects, whereas diagonal arrows represent cross-lagged effects. Path coefficients are standardized. Gender and school types were controlled as covariates.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

CLPM framework.

3.4. Random-intercept cross-lagged panel model

Table 2 presents the model fits of the constrained and unconstrained RI-CLPM models. The intraclass correlation coefficients were 0.358 for cyberbullying, 0.308 for MD, and 0.224 for NOBAGs. Differences between subjects therefore accounted for 22.4%–35.8% of the variance in the study variables, while the remaining 64.2%–77.6% of the variance was explained by within-person variations over time. The RI-CLPM was therefore sufficient to detect within-person changes (i.e., no less than 10%) over time (Hamaker et al., 2015).

The unconstrained RI-CLPM model met the conventional criteria for excellent fit ($\chi^2(14) = 22.881$, $p > 0.05$; CFI = 0.995; TLI = 0.982; RMSEA = 0.026, 90% CI = 0.000–0.044; SRMR = 0.016), while the constrained RI-CLPM failed to meet the criteria for even acceptable fit ($\chi^2(35) = 371.371$, $p < 0.001$; CFI = 0.828; TLI = 0.734; RMSEA = 0.100, 90% CI = 0.091–0.109; SRMR = 0.111). The model fits improved when autoregressive and cross-lagged effects were freely estimated, and the unconstrained model was therefore adopted for subsequent analyses.

Fig. 2 shows the standardized path estimates of the unconstrained RI-CLPM model. The between-person intercepts of cyberbullying, NOBAGs, and MD were all positively correlated ($p < 0.001$), and there were significant and positive autoregressive effects for cyberbullying and NOBAGs. Cyberbullying at T_n predicted both NOBAGs at T_{n+1} ($\beta_{\text{Cyberbullying}, T1 \rightarrow \text{NOBAGs}, T2} = 0.109$, $p < 0.05$; $\beta_{\text{Cyberbullying}, T2 \rightarrow \text{NOBAGs}, T3} = 0.108$, $p < 0.05$) and MD at T_{n+1} ($\beta_{\text{Cyberbullying}, T2 \rightarrow \text{MD}, T3} = 0.116$, $p <$

0.05). H1 and H3 were therefore partially supported at the within-person level. NOBAGs at T2 also significantly predicted MD at T3 ($\beta_{\text{NOBAGs}, T2 \rightarrow \text{MD}, T3} = 0.161$, $p < 0.01$). H2 was thus also partially supported at the within-person level. These results suggest that the subjects' cyberbullying behaviors have a unidirectional relationship with NOBAGs and MD over time.

Longitudinal mediation analysis was then performed to test the mediating role of NOBAGs. The results show that the indirect effect of cyberbullying at T1 on MD at T3 via NOBAGs at T2 was not significant ($\beta = 0.018$, $p = 0.059$, 95% CI = −0.001–0.036), so H4 was not supported by the RI-CLPM: NOBAGs did not mediate the longitudinal relationship between cyberbullying and MD when between-person and within-person variations were separated.

4. Discussion

While the predictive roles of NOBAGs and MD in the development of cyberbullying have been shown in previous studies, few longitudinal studies have examined how cyberbullying behaviors lead to a deterioration in individual moral standards and greater acceptance of aggression in opposition to social norms. This study tested the temporal dynamic relationships between cyberbullying, NOBAGs, and MD using both a CLPM and RI-CLPM in a sample of primary school children in Hong Kong. In the CLPM, direct reciprocal relationships were found between cyberbullying and NOBAGs and between NOBAGs and MD but not between cyberbullying and MD, with NOBAGs mediating the longitudinal relationship between cyberbullying and MD. However, by

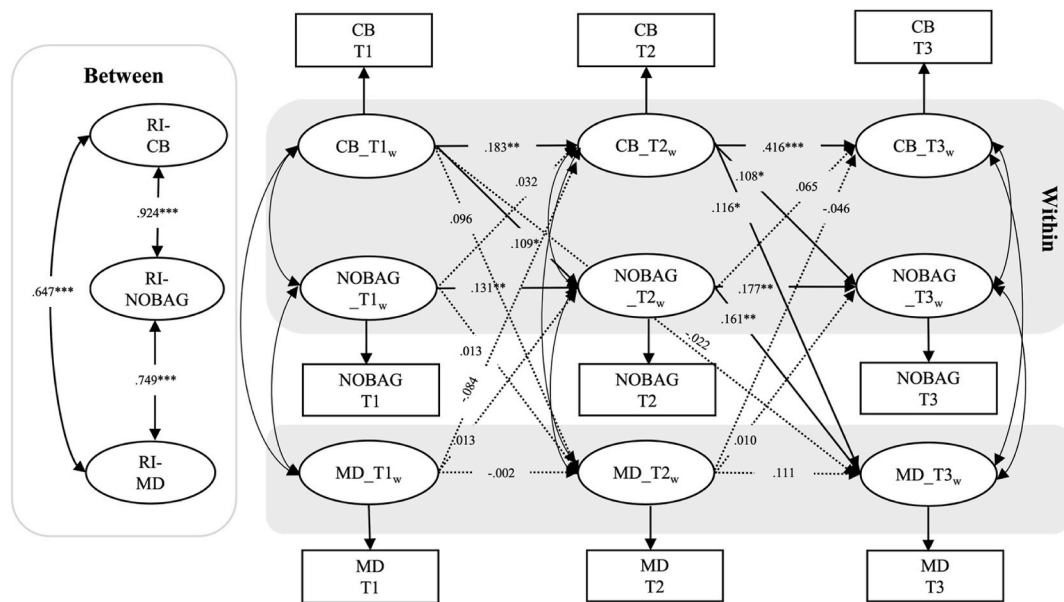


Fig. 2. Random-intercept cross-lagged panel model of cyberbullying, normative beliefs about aggression, and moral disengagement.

Note. CB = Cyberbullying; MD = Moral disengagement; NOBAG = Normative beliefs about aggression. Squares indicate observed variables. Ellipses denote latent variables. Factor loadings for both between-person and within-person variables were fixed at 1. Solid arrows depict significant pathways, while dashed arrows signify non-significant ones. Correlations are illustrated with double-headed arrows. Horizontal arrows represent autoregressive effects, and diagonal arrows depict cross-lagged effects. The nine within-person latent variables (CB_T1w, CB_T2w, CB_T3w, NOBAG_T1w, NOBAG_T2w, NOBAG_T3w, MD_T1w, MD_T2w, and MD_T3w) represent within-person variances across Time 1 to 3. All path coefficients are standardized. Gender and school types were controlled as covariates.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

partitioning between-person stability from within-person variances, the RI-CLPM found more nuanced within-person dynamics. After controlling for stable individual heterogeneity, only cyberbullying significantly and positively predicted subsequent NOBAGs and MD, and no mediating effects of NOBAGs were found.

4.1. Between-person relationships between cyberbullying, NOBAGs, and MD

Consistent with existing evidence (e.g., Bandura, 2016; Falla et al., 2023; Kowalski et al., 2014), positive associations were observed between cyberbullying, MD, and NOBAGs at the between-person level in the RI-CLPM. Between-person effects reflect stable trait-like differences between individuals that persist over time, in contrast to temporary situational variations (Hamaker et al., 2015). The current findings, therefore, suggest that children who engage more in cyberbullying also have greater NOBAGs and MD; this is consistent with social cognitive theory, which posits a vicious circle between inhumane behaviors and moral deviancy.

In social cognitive theory, when a person is developing their moral self, there is an evaluative self-regulatory process in which both internalized moral standards and external environments influence how they assess their own behaviors (Bandura, 2016). During this process, MD can be selectively triggered in contexts in which bullying is performed; such a moral deviation allows a person to temporarily disengage from moral self-regulation, display fewer moral emotions, and reject self-sanctions (Hymel et al., 2005; Sticca & Perren, 2015). Thus, the significant between-person relationship found between cyberbullying and NOBAGs in the current study may reflect a significant impact on the development of personal NOBAGs of both external social influences and specific personal experiences. Specifically, contextual factors such as social norms may facilitate the emergence of NOBAGs (Ang et al., 2011; Shi et al., 2021; H. Zhou, Wei, et al., 2023), while individuals' observations of their own aggressive behaviors may reinforce and solidify an aggressive orientation (Orue et al., 2023).

4.2. Cyberbullying leading to deteriorating NOBAGs and MD

The within-person—that is, state-like—findings present a more complex temporal dynamic. More cyberbullying behaviors may encourage children to believe that cyber aggression can be acceptable in certain situations and to increasingly rely on MD mechanisms to justify their behaviors. Falla et al. (2023) described how cyberbullying behaviors contribute to the development of immorality among young people through a gradual process of desensitization to aggression, and ample additional evidence suggests that cyberbullying can encourage individuals to abandon or lower their moral standards and avoid recognizing their own violations of moral norms (Falla et al., 2021). Through repetitive aggressive behaviors, they gradually internalize the idea that aggressive interactions can be justified and accepted as a means of communication, leading to the habitual and unreflective normalization of unethical actions in the long term (Bandura et al., 1996; Bandura, 2016). Aggressive behaviors may also generate rewards that enhance the perpetrator's sense of autonomy—especially among young people—or satisfy popularity needs within peer groups, which in turn may make it easier for them to form self-serving moral justifications and thus weaken their own moral constraints (Romera et al., 2021). Consequently, widespread cyberbullying can fuel a trend of escalating MD and may embed immoral justifications into children's moral and cognitive schemas, potentially leading to more serious problems.

A significant pathway from NOBAGs to cyberbullying was also identified in the CLPM, but this effect became non-significant at the within-person level in the RI-CLPM. Our hypothesis regarding the bidirectional relationship between NOBAGs and cyberbullying was therefore not fully supported, which fits with prior research that has been inconsistent in finding a link from NOBAGs to cyberbullying; indeed, some studies—especially those involving young people—have found that NOBAGs did not predict subsequent aggressive behaviors at all (e.g., Krahé & Busching, 2014; Orue et al., 2023). The path from NOBAGs to cyberbullying found in the CLPM and at the between-person level in the RI-CLPM suggests that NOBAGs are primarily a stable individual characteristic. Because they are established early in childhood

and stabilize over time (Sticca & Perren, 2015), NOBAGs remain relatively consistent within individuals across time and are culturally reinforced, particularly in collectivist cultures (e.g., Y. Li et al., 2012; Shi et al., 2021; F. Yang, Jin, et al., 2022). However, we did find the inverse: that individual cyberbullying experiences exerted a significant influence on within-person changes in NOBAGs.

Notably, neither the CLPM nor the within-person results found that MD promotes cyberbullying, contrary to our hypothesis. Although a number of studies have argued that MD is a predictor of cyberbullying (e.g., Lo Cricchio et al., 2020; Paciello et al., 2020), recent RI-CLPM studies have found that within-person increases in MD predict subsequent increases in reactive and general aggression but not in proactive aggression (Lin et al., 2026; Lin & Xia, 2026). Considering that cyberbullying is often conceptualized as primarily proactive aggression (Calvete et al., 2010), those findings may help to explain the non-significant within-person effects observed in our RI-CLPM. Furthermore, Lin et al. (2026) suggested that, when responding to MD assessments, young Chinese people may focus on justifying reactive aggression rather than proactive aggression, influenced by Chinese cultural norms that emphasize tit-for-tat responses. MD may therefore be more closely associated with reactive, emotionally driven aggression than with goal-oriented aggressive behaviors such as cyberbullying.

4.3. Mediating effects of NOBAGs on the relationship between cyberbullying and MD

NOBAGs functioned as a mediator between cyberbullying and MD in the CLPM analysis, consistent with our hypothesis, but this effect likely reflects spurious mediation arising from shared individual differences because it was not replicated at the within-person level in the RI-CLPM. That is, after controlling for trait-like differences, variations in cyberbullying were not associated with subsequent changes in MD via changes in NOBAGs. Cyberbullies therefore appear to activate proximal MD mechanisms to justify their own actions, without detectable cross-wave changes in NOBAGs as an intermediary process, and this discrepancy may thus indicate an immediate moral rationalization process that children use internally when perpetrating cyberbullying.

The absence of a within-person mediation effect also suggests that the cognitive pathway from cyberbullying through NOBAGs to MD may be more related to stable between-person differences than something that changes within individuals over time. Such stable differences may stem from enduring individual traits (e.g., self-control, psychopathy; H. Li et al., 2023) or broader social-environmental factors (e.g., family support, peer environments; Miao & Li, 2024) at a more macro level than momentary intra-individual changes. Intense academic pressure and competitive educational environments (Chyu & Chen, 2022; Y. Li et al., 2012)—which are a feature of China and of East Asia more broadly—may create enduring individual differences in how children deal with stress and interpersonal conflict. Some may develop stable patterns of externalizing behaviors—including cyberbullying and moral-cognitive distortions—as trait-like coping mechanisms, and early, sustained, and systematic environmental and educational interventions are therefore crucial to foster moral self-regulation and reduce such trait-like vulnerabilities associated with aggressive behaviors.

Another possible explanation for the differences between the CLPM and RI-CLPM findings relates to differences in the methodologies themselves. CLPMs do not separate stable between-person differences from within-person processes, which may prevent accurate capture of causal changes (Hamaker et al., 2015), while the isolation of within-person variations in the RI-CLPM may provide a more accurate test of longitudinal associations. Many psychological variables are also highly stable over time, and when previous levels are taken into account in longitudinal models, a large portion of shared variance tends to be removed (Adachi & Willoughby, 2015; J. Zhou, Zhao, et al., 2023). In the RI-CLPM, this reduction is deepened by the exclusion of stable between-person variance (J. Zhou, Zhao, et al., 2023), and the

remaining within-person variability thus tends to be limited, which may partly explain why the cross-lagged effects are attenuated or statistically non-significant.

4.4. Limitations and implications

Several limitations need to be considered. First, this study was conducted with a sample of children from Hong Kong, so the findings may not be generalizable to other populations. It is thus recommended that the relationships between cyberbullying, NOBAGs, and MD be validated with samples in multiple age groups and in more varied cultural contexts.

Second, the study variables were all assessed using self-report instruments. Relying on the same informant for multiple constructs can create biases due to common method variance (Podsakoff et al., 2012), and although these biases can be reduced by separating predictor and criterion variables temporally (Podsakoff et al., 2003), they may still be present in longitudinal studies. Personal bias, such as social desirability, may also arise in self-report data—particularly in research involving socially sensitive topics such as bullying—which can then compromise the validity of the findings. Future studies should therefore adopt multi-source designs and incorporate data from diverse informants (e.g., teachers, parents) to provide a more comprehensive assessment of the key constructs.

Third, this study did not assess external factors (e.g., social support, parenting styles) that may influence the development of cyberbullying or the relationships between the study variables. Future research should therefore consider incorporating these factors.

Fourth, the RI-CLPM is inherently more conservative than the CLPM in detecting indirect effects because it partitions variance and has less statistical power for mediation pathways that require multiple significant paths (Burns et al., 2020). Larger sample sizes may therefore be needed to achieve sufficient power if the RI-CLPM is used in future studies (Hamaker, 2018).

Nevertheless, this study extends the application of social cognitive theory and social cognitive information processing models to cyberspace and increases our understanding of moral development in the age of pervasive digital interactions. The study also shows—by uncovering longitudinal, bidirectional relationships between cyberbullying, NOBAGs, and MD—that, while cyberbullying is influenced by internal moral cognition, it also reshapes those same cognitions and beliefs during repeated interactions with the social environment.

Methodologically, the combined use of CLPM and RI-CLPM allows for a more nuanced and differentiated understanding of longitudinal associations. The CLPM results helped to establish the presence of dynamic relationships between cyberbullying, NOBAGs, and MD, while the RI-CLPM clarified which of those associations could be attributed to intra-individual change. Notably, evaluating the longitudinal development of constructs or their interconnections over time using a single analytical model can be difficult, and the interpretations and implications of these results should therefore be considered within the appropriate methodological context.

Finally, this study has practical implications. Support systems should be established to provide timely mental and emotional support—such as counseling services—to children who are involved in cyberbullying activities, either as perpetrators or as victims. Preventative measures and stronger supervision are also needed to prevent or interrupt the vicious cycle between cyberbullying and MD. Interventions could include off-line extracurricular activities, programs to encourage responsible internet use and foster awareness of the consequences of online aggression, and training courses to improve children's digital literacy, emotional management, and conflict resolution.

5. Conclusion

The reciprocal effects of cyberbullying, NOBAGs, and MD and the

mediating role of NOBAGs in a sample of Chinese children were examined in this study using a traditional CLPM analysis, with an RI-CLPM analysis offering a complementary perspective for testing the causal mechanisms. After separating within-person changes from between-person differences, cyberbullying predicted the development of subsequent MD directly rather than through NOBAGs, while the between-person relationships remained significant. In contrast to previous evidence on the predictive role of MD in cyberbullying (e.g., Lo Cricchio et al., 2020; Paciello et al., 2020), an inverse reinforcement pattern was found in which perpetrating cyber aggression triggers both changes in normative beliefs and moral rationalization. The findings deepen our knowledge of the adverse effects of cyberbullying behaviors on cognitive development among children and provide insights into developing targeted interventions.

CRediT authorship contribution statement

Xintong Zhang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xiang Li:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis.

Ethical statement

The project has been approved by the Human Subjects Ethics Subcommittee (HSESC) of The Hong Kong Polytechnic University (HSESC Reference Number: HSEARS20190321001).

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Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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